

Which Machine Learning Models Are Actually Used in Industry?

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Abstract

Recent advances in artificial intelligence have generated significant interest in deep learning and Large Language Models (LLMs). However, the models that attract the most research attention are not always the models most commonly used in industry. This report reviews evidence from industry surveys, technical reports, and company analyses to identify the machine learning models that are most widely deployed in real-world applications. The findings indicate that classical machine learning approaches, particularly regression models and tree-based ensemble methods, remain dominant for structured data problems. At the same time, deep learning and LLM-based systems are rapidly expanding in domains involving unstructured data such as text, images, and audio. The report concludes with a prediction regarding how the distribution of machine learning models is likely to evolve during the next decade.

1 Introduction

Machine learning has become an essential technology across modern industries. Organizations use machine learning systems for customer analytics, fraud detection, recommendation systems, predictive maintenance, healthcare diagnostics, financial forecasting, and many other applications. During the last decade, advances in deep learning have significantly improved performance in tasks involving images, speech, and natural language processing. More recently, the success of Large Language Models (LLMs) such as ChatGPT has further accelerated interest in artificial intelligence.

Despite this excitement, industrial adoption is influenced by factors beyond predictive accuracy. Businesses must consider interpretability, computational cost, deployment complexity, maintenance requirements, and regulatory constraints. Consequently, the most advanced model is not necessarily the most practical model. This report investigates which machine learning models are currently most widely used in industry and examines future trends in industrial machine learning.

2 Most Commonly Used Models in Industry

2.1 Linear and Logistic Regression

Linear and logistic regression remain among the most widely used machine learning techniques in industry. According to Kaggle’s State of Data Science and Machine Learning Survey [1], logistic regression was one of the most frequently used algorithms among practicing data scientists. Similarly, industry surveys conducted by Rexer Analytics consistently identify regression methods as one of the most commonly applied analytical tools [2].

The popularity of regression-based models stems from several advantages. First, they are highly interpretable, allowing organizations to understand the relationship between input variables and predictions. Second, they are computationally efficient and can be trained quickly even on large

datasets. Finally, they perform well in many structured-data applications such as credit scoring, risk assessment, customer churn prediction, and business analytics.

For industries subject to regulatory requirements, such as finance and healthcare, interpretability is often essential. In these settings, regression models remain attractive because they provide explanations that can be communicated to stakeholders and regulators.

2.2 Decision Trees and Ensemble Methods

Decision trees and tree-based ensemble methods are arguably the most important family of machine learning models used in modern industry. Rexer Analytics reports that decision trees have consistently ranked among the most widely used machine learning techniques across multiple survey years [2].

Although individual decision trees are simple and interpretable, their ensemble versions provide substantially higher predictive performance. Popular ensemble methods include Random Forests, Gradient Boosting Machines (GBMs), XGBoost, LightGBM, and CatBoost. These algorithms combine the predictions of multiple trees to improve accuracy and reduce overfitting.

Tree-based ensembles are particularly effective for structured and tabular datasets. In many industrial applications, they outperform deep neural networks while requiring less training data and computational power. Microsoft’s LightGBM project highlights the importance of gradient-boosted decision trees in modern machine learning pipelines [4]. Consequently, tree-based methods have become the standard solution for many business-oriented machine learning tasks including forecasting, recommendation, risk prediction, and customer analytics.

2.3 Clustering and Unsupervised Learning

Unsupervised learning methods continue to play an important role in industry. Algorithms such as K-Means clustering, hierarchical clustering, and anomaly detection techniques are widely used for customer segmentation, cybersecurity, fraud detection, and exploratory data analysis.

Unlike supervised learning methods, clustering algorithms do not require labeled data. This characteristic is particularly valuable because many organizations possess large volumes of unlabeled information. According to industry surveys, clustering consistently remains among the most frequently used analytical methods [2, 3].

Although clustering algorithms receive less attention than deep learning, they continue to provide substantial business value by uncovering hidden patterns and supporting data-driven decision-making.

2.4 Deep Learning Models

Deep learning has revolutionized fields involving unstructured data. Convolutional Neural Networks (CNNs) dominate computer vision applications such as image classification, object detection, and medical image analysis. Transformer-based architectures have become the leading approach in natural language processing and speech recognition.

Technology companies such as Google, Microsoft, Meta, and OpenAI rely heavily on deep learning systems for search, recommendation engines, translation services, and generative AI applications. Nevertheless, deep learning is not universally dominant across all industrial applications.

Deep neural networks often require large datasets, expensive computational resources, and specialized expertise. As a result, organizations frequently prefer simpler machine learning methods when dealing with structured business data. Recent analyses of machine learning

competitions continue to show that gradient-boosted decision trees often outperform deep neural networks on tabular datasets [4].

2.5 Large Language Models

The emergence of Large Language Models represents one of the most significant developments in artificial intelligence during the past few years. Models based on the Transformer architecture introduced by Vaswani et al. [6] have enabled applications such as conversational agents, document analysis, code generation, search augmentation, and knowledge management.

Industry reports indicate rapid enterprise adoption of LLM technologies. Databricks reports that natural language processing is among the fastest-growing AI application areas and that organizations are increasingly investing in generative AI capabilities [5]. However, LLMs are typically deployed alongside existing machine learning systems rather than replacing them entirely. In many real-world applications, classical machine learning models continue to perform prediction and decision-making tasks while LLMs provide natural language interfaces and reasoning capabilities.

3 Current Industry Landscape

Based on the reviewed evidence, the most widely used machine learning models in industry can be grouped into the following categories:

1. Linear and Logistic Regression
2. Decision Trees and Tree-Based Ensembles
3. Clustering and Unsupervised Learning Methods
4. Deep Learning Models (CNNs and Transformers)
5. Large Language Models (rapidly growing adoption)

Among these categories, regression models and tree-based ensemble methods currently remain the dominant solutions for structured-data applications. Deep learning and LLM-based approaches are expanding rapidly but are primarily concentrated in domains involving text, images, audio, and other forms of unstructured data.

4 Prediction for the Future

Over the next 5–10 years, classical machine learning models are unlikely to disappear. Instead, they will continue to play a central role in applications involving structured and tabular data. Industries such as banking, insurance, manufacturing, energy systems, and operations management often require models that are interpretable, computationally efficient, and easy to audit. For these reasons, logistic regression, random forests, gradient boosting methods, and related techniques are expected to remain widely used. Furthermore, many organizations do not possess the massive datasets required to fully exploit advanced deep learning architectures, making classical approaches the most practical solution.

At the same time, deep learning and LLM-based systems are expected to capture a significantly larger share of industrial machine learning deployments. The strongest growth will likely occur in natural language processing, computer vision, speech processing, robotics, autonomous systems, and multimodal AI applications. Improvements in hardware, cloud computing infrastructure, and open-source foundation models will lower the barriers to adoption. As a result,

organizations that previously relied exclusively on traditional machine learning methods will increasingly integrate deep learning capabilities into their workflows.

Rather than one model family replacing another, the most likely future scenario is a hybrid ecosystem. Classical machine learning methods will continue to dominate structured-data problems, while deep learning and LLMs will dominate applications involving language, images, audio, and complex reasoning tasks. For example, a financial institution may use gradient-boosted decision trees to estimate credit risk while simultaneously using LLMs to analyze documents and communicate with customers. Consequently, future industrial AI systems will likely combine multiple model families, leveraging the strengths of each approach.

5 Conclusion

The evidence reviewed in this report suggests that classical machine learning models remain the most widely used algorithms in industry despite the growing popularity of deep learning and generative AI. Regression models, decision trees, random forests, and gradient-boosting techniques continue to dominate structured-data applications because they provide strong performance, interpretability, and operational simplicity.

However, the influence of deep learning and LLMs is increasing rapidly. These technologies are transforming domains that involve text, images, speech, and other forms of unstructured data. Over the next decade, the machine learning landscape is expected to become increasingly diverse, with organizations combining classical methods and modern AI systems to address different business requirements. Therefore, the future of industrial machine learning is likely to be characterized by integration rather than replacement.

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